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A three dimensional model of muckpile formation and grade boundary movement in open pit blasting

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Summary

Formulation and case studies of a three dimensional kinematic model are presented. The *in situ* overburden geometry can be simulated accurately and various initiation patterns of blasts can be modelled. The overburden geometry, hole patterns and explosive distribution are all explicit model inputs. Because the effect of explosive properties, rock mass condition and inter-row delay are very difficult to measure in terms of blast performance, these are represented in the model by control parameters which are left for calibration using field data. The output of the model is a three dimensional muckpile shape of any cross section and a contour map of grade distribution within the muckpile. Two case studies are presented which have shown that the model is a valuable tool for optimizing production blasting as well as for controlling grade dilution during blasting.

Keywords: Blasting; three dimensional model; muckpile shape; grade control; explosive performance

Introduction

Modelling of blasting has tended to be concentrated on the fragmentation process and modelling of muckpile formation has received relatively scant attention in the literature. At present, there are few documented case studies of successful applications of modelling to predict muckpile shapes and there are no three dimensional models of muckpile formation. Yet this aspect of the blasting process is particularly important in open pit mining.

Muckpile shape and productivity

Measures of blast effectiveness in surface mining operations are different from those used in underground operations. In the underground situation, particle size has a great influence on productivity. In surface mining operation, however, the ease with which a muckpile may be dug is within broad limits largely independent of fragment size distribution, since the equipment used to move the broken rock is generally much larger than in underground mines. In such cases, muckpile shape strongly influences mine productivity. In open pit mining situations, a suitable muckpile shape after blasting is required for a selected piece of

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digging equipment. A front-end loader is best suited to low, flat-lying, loose muckpiles, as most front-end loaders only generate a fairly small break-out force with the bucket and can work very efficiently over a large area. In contrast, a rope shovel lacks mobility, generates a much larger bucket break-out force and the bucket is filled in a vertical filling motion. This makes the rope shovel ideally suited to well fragmented, steep faces (Bulow, 1988). In some strip mining situations, throw blasting is designed to cast as much as possible of the spoil across the pit with explosives. Evaluation of the technical and economic aspects of the method has indicated that considerable benefits can be achieved, provided that the environmental impact is acceptable. It is necessary to optimize the design of the throw blast in terms of maximizing the throw percentage under the necessary constraints. For dragline operations in some strip mining situations, the muckpile shape is of more interest than the percentage of throw across the pit. If the dragline operation uses a conventional bridge, the preferred working level of the dragline is at the top of the new highwall. This dig working level must be maintained, so that the dragline may sit over the key cut and little preparation of the dragline pad should be required.

Grade control during blasting

In many open cut operations, in particular gold, nickel and iron ore mines, the delineation of regions of differing grade and the ability to mine those regions separately from within the same blast pattern is critical. Delineation of grade regions prior to blasting is usually known to an accuracy of within a few metres (at least in plan) since blastholes are assayed routinely. After the blast has occurred, grade boundaries very rarely follow the pre-blast regions since significant displacements occur. Unless there are marked colour differences between regions, large errors can be incurred in the laying out of post-blast digging regions. Thus prediction of grade region boundary movement and subsequent dilution is of interest to many open cut mines. Such a predictive tool would enable blast design changes to be made to reduce mixing of certain boundaries.

Current modelling approaches and their difficulties

Most existing models of blasting are dynamic, namely, the model calculations are based on the fundamental physics to analyse the detailed interactions between explosive and rock overburden (Cherry, 1967 and Schamaun, 1983). In fact, details of the complex dynamic mechanisms of rock blasting are still far from clear at present because of the difficulties in measuring the behaviour of the explosive and the response of the rock mass during blasting. In using the dynamic method to model the blasting process many uncertain parameters will be involved. This method is normally complex to implement and takes a large amount of computer time. This forms a major impediment to the simulations of real production blasts.

Modelling approach selected

The model presented in this paper is based on a simpler 'kinematic' approach to modelling muckpile formation. It is an extension of the two-dimensional kinematic model reported by Yang and Kavetsky (1989). The method does not calculate the velocitiy from the acceleration process of the fragmented rock block. The model is simplified by determining major trends and macro-phenomenon and includes both fundamental and empirical aspects. This

modelling approach has the advantage of relative simplicity, while still reflecting the essential features of the blasting displacement process. Because of the relatively simple implementation, the control parameters of the model can be calibrated against field data in a straight forward manner and then used for predictions at the same site.

The geometry of the blast design includes burden, spacing, drilling pattern, hole inclination, bench height, free face configuration and boundary confinement. Due to the limitations of the previous modelling approach, most existing models of blasting are two dimensional and the detailed geometry of blasts is often ignored or over simplified (Cherry, 1967, Schamaun, 1983, Harries, 1987 and Hyde and Faureau, 1986). However, the model presented in this paper can simulate the *in situ* overburden geometry accurately. Various initiation patterns of blasts can also be modelled.

The geometry of the blast design is a major aspect of blasting and has a pronounced effect in practice on both rock fragmentation and fragment motion. For example, if the burden of a hole is too small to provide suitable confinement to the explosive gas pressure, much of the gas energy will be lost to the atmosphere before it is able to contribute fully to fragmentation or fragment motion. If a very big toe burden is adopted, unacceptable toe or poor fragment displacement will occur. It is also significant to note that even with the same blast geometry, different firing patterns or sequences often result in quite different burden movement.

Three basic assumptions are incorporated in the model. Firstly, when the rock begins to move it is assumed to have been already fragmented, either by the stress wave or by geological jointing, or both. Secondly, it is assumed that the rock fragments move with an initial velocity as projectiles (Lemesh and Pozdynakov 1971). Thirdly, it is assumed that the initial velocity of each rock fragment is determined by two factors, namely, the energy imparted to the fragment from the energy source and the amount of resistance to movement received from surrounding fragment blocks. For example, a fragment block close to a free face is subjected to less resistance than those further into the rock mass.

Model calculations

Model calculations can be identified as occurring in two parts. The first is a geometrical calculation, which includes: (1) accurate description of the boundaries of the overburden and division of the overburden into computational blocks; (2) calculation of grade distribution and (3) calculation of the muckpile. A computational block may contain a number of adjacent rock fragments which have nearly the same velocity. In the following discussion, the computational block is simply referred to as a block. The second part is the calculation of block velocity, based on the block coordinates and the explosive distribution.

Geometry of the model

Plane determined by a row of blast holes. A detonating row of blast holes is defined as those holes detonated within a small time period and spaced within a certain distance. Open cut blasts are usually fired row by row. Each detonating row of blast holes defines a plane, which can be represented by an equation calculated according to the coordinates of the hole collars and the directions of the holes in the row.

Overburden boundaries represented as planes. Planes are used to represent boundaries of the overburden, namely, free faces, side boundaries and the back boundaries, etc. as seen in Fig. 1. The equations of these planes are calculated according to the orientation and location of overburden boundaries. For a 'V' pattern blast, a vertical central plane is used to separate the left side and the right side of the blast.

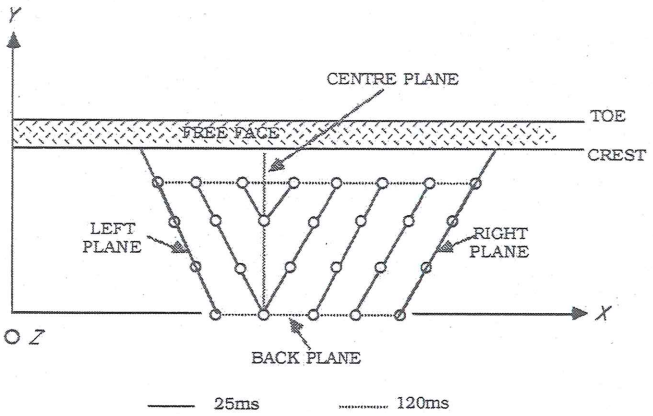


Fig. 1. Overburden boundaries represented as planes

Dividing overburden into computational blocks. The planes defined by the detonating rows divide the overburden of a blast into sub-burdens (i.e. the overburden between two successive detonating rows of blast holes, as shown in Fig. 2a). Figs 2a and 2b show that a sub-burden is divided into computational blocks. It is considered that blocks within each sub-burden are moved by explosive energy from holes in the row immediately behind it.

The coordinates of all blocks are calculated, for use in the velocity calculations for the blocks. The volume of each block is also determined for the assessment of the muckpile shape and size. The number of blocks in the vertical direction (N_v) for the sub-burden can be determined according to the height of the sub-burden (H_s) divided by the nominal block size (D). Similarly, the number of blocks in the horizontal direction (N_h) can be calculated with the average of the crest and toe burdens of the sub-burden divided by the nominal block size. The sub-burden is considered as a superimposition of N_v 'slices' with thickness D_v (H_s divided by N_v). The ith slice in the sub-burden is displayed, in Fig. 2a. The width L_s and L_e of the slice in Fig. 2b can be calculated from the equations of the boundary planes. The slice is further considered as an assembly of 'bars' made up of adjacent blocks. The width of a bar is:

$$D_h = \frac{L_s + L_e}{2N_h} \tag{1}$$

Figure 2b shows the jth bar in the ith slice. It is obvious that the coordinates (X_s, Y_s) and (X_e, Y_e) can be determined from the equations of the planes. Therefore, the length of the jth bar, the number of blocks in the bar and the length of a block, respectively, are:

$$L_b = \sqrt{(X_e - X_s)^2 + (Y_e - Y_s)^2}; \quad N_r = \frac{L_b}{D} \quad \text{and} \quad D_r = \frac{L_b}{N_r} \tag{2}$$

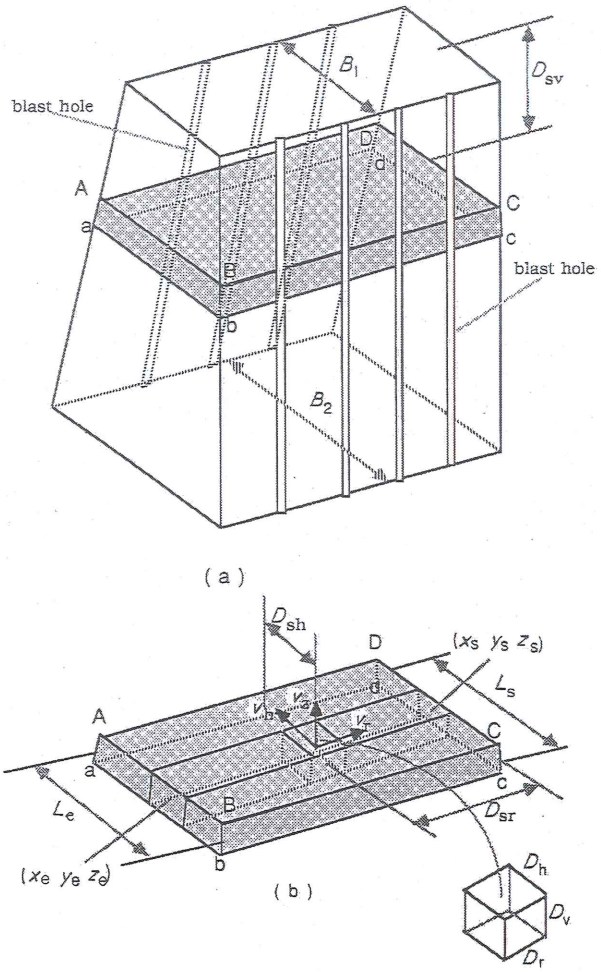


Fig. 2. Dividing overburden into computation blocks. (a) A sub-burden considered as a superimposition of 'slices', (b) a 'slice' considered as an assembly of 'bars' made up of adjacent blocks

The directional cosines of the bar are:

$$\xi_x = \frac{X_e - X_s}{L_b}; \quad \xi_y = \frac{Y_e - Y_s}{L_b}; \quad \text{and} \quad \xi_z = 0 \tag{3}$$

Consequently, the kth block in the jth bar of the ith slice, as shown in Fig. 2b has the following coordinates:

$$X_{i,j,k} = X_s + \xi_x D_r (k - 0.5) \tag{4}$$

$$Y_{i,j,k} = Y_s + \xi_y D_r (k - 0.5) \tag{5}$$

$$Z_{i,j,k} = Z_s \tag{6}$$

where, $i = 1, \ldots, N_v$; $j = 1, \ldots N_h$ and $k = 1, \ldots N_r$.

The volume of the block is

$$V = D_v D_r D_h \tag{7}$$

Grade distribution calculation. Based on drilling assays, the overburden can be defined as regions with different grade values. Each block can then be assigned a grade value based on the region where the block was originally located. The distribution of the valuable mineral in the muckpile determined by block locations after blasting gives the grade distribution within the muckpile.

Block landing and muckpile shape. It is assumed that the first blocks to move are those closest to the free face. This assumption is based on high speed photography analysis, in particular that detailed by Lemesh and Pozdynakov (1971).

When a block lands on the bench floor, it remains at that point without further movement. When a block hits an obstacle, such as earlier arriving blocks, its flight is terminated and it is brought to rest at one block size in front of the obstacle. The muckpile height is represented as two dimensional square grids with the grid size equal to the nominal block size. If a block with a volume V lands within a grid square, the volume of the block is distributed to the local muckpile height at the grid locations.

Block velocity

The velocity imparted to any block is assumed to be due only to the charge (or charges) in the row immediately behind the block. As energy from an explosive segment radiates spherically from the source, geometric spreading reduces the energy per unit area according to the inverse of the square of the distance from the source. In addition energy is also absorbed by rock fracture and other attenuation mechanisms during propagation through rock mass.

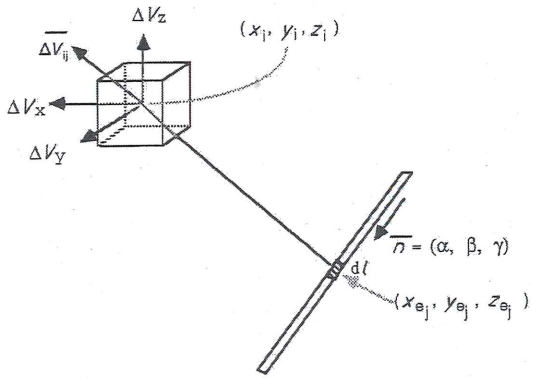


Fig. 3. Notation of velocity calculation from a charge column

It is assumed that the change in speed of a block due to any explosive segment is directly related to the energy contributed to the block from the explosive segment. Therefore, the increment in speed of block (X_i, Y_i, Z_i) due to explosive segment d1 at (X_{ej}, Y_{ej}, Z_{ej}) is given by Equation 8 and illustrated in Fig. 3:

$$|dV_{ij}| = P_e \frac{d1}{r_{ij}^{2P_r - 1}} \tag{8}$$

where P_e and P_r are 'explosive' and 'rock' parameters. The components of dV_{ij} in the coordinate directions are given by Equations 9–11:

$$dV_x = |dV_{ij}| \frac{X_i - X_{ej}}{r_{ij}} \tag{9}$$

$$dV_y = |dV_{ij}| \frac{Y_i - Y_{ej}}{r_{ij}} \tag{10}$$

$$dV_z = |dV_{ij}| \frac{Z_i - Z_{ej}}{r_{ij}} \tag{11}$$

where

$$r_{ij} = \sqrt{(X_i - X_{ej})^2 + (Y_i - Y_{ej})^2 + (Z_i - Z_{ej})^2}$$

The component velocities of the *i*th block due to all relevant charges are produced by substituting Equation 8 into Equations 9–11, integrating over the charge length L_j, and summing for all relevant charges (m):

$$Vx_i = \sum_{j=1}^{m} \int_0^{L_j} P_e \frac{(X_i - X_{ej})\, d1}{[(X_i - X_{ej})^2 + (Y_i - Y_{ej})^2 + (Z_i - Z_{ej})^2]^{P_r}} \tag{12}$$

$$Vy_i = \sum_{j=1}^{m} \int_0^{L_j} P_e \frac{(Y_i - Y_{ej})\, d1}{[(X_i - X_{ej})^2 + (Y_i - Y_{ej})^2 + (Z_i - Z_{ej})^2]^{P_r}} \tag{13}$$

$$Vz_i = \sum_{j=1}^{m} \int_0^{L_j} P_e \frac{(Z_i - Z_{ej})\, d1}{[(X_i - X_{ej})^2 + (Y_i - Y_{ej})^2 + (Z_i - Z_{ej})^2]^{P_r}} \tag{14}$$

Confinement for each block. In order to take into account the variation in confinement for each block, a factor is used to modify the resultant velocities of blocks. The block velocity is resolved into three components, as seen in Fig. 2b, i.e. the resultant horizontal velocity along the bar's long dimension (V_r) and the resultant horizontal velocity normal to the bar's long dimension (V_h) and the resultant vertical velocity (V_z). Based on the assumption that a block close to a free face suffers less resistance than those further into the rock mass, velocity components of a block are modified as follows:

$$V_h^c = V_h \exp(-C_1 D_{sh}) \tag{15}$$

$$V_r^c = V_r \exp(-C_1 D_{sr}) \tag{16}$$

$$V_z^c = V_z \exp(-C_1 D_{sv}) \tag{17}$$

where D_{sh}, as shown in Fig. 2b, is the distance (in metres) from the block to the free face (for

the first row) or to the new free face, (created by the immediately preceding row) in the direction of the velocity V_h. D_{sr} is the distance (in metres) from the block to a free face in the direction of the velocity V_r (refer to Fig. 2b) and D_{sv} is the vertical distance from the block to the surface of the blast (refer to Fig. 2a). If the blast is completely confined in the V_r direction, D_{sr} is set to infinity. The parameter C_1 reflects the resistance to motion caused by the surrounding blocks. In model calculations, C_1 is taken to be a constant with an empirically determined value of 0.2.

Interrow delay. In practice, the movement of overburden rock is strongly affected by delay time. For a given number of rows, as the delay between rows decreases, the vertical component tends to increase relative to the horizontal component. The model takes this into account by multiplying the initial horizontal velocity component (V_h^c) (normal to the bar's long dimensional) by a delay factor, i.e.

$$V_h^d = V_h^c \exp(-P_d n/T_d) \tag{18}$$

Where,

n is the number of rows in front of the block.

V_h^d the horizontal velocity (normal to the bar's long dimension) of the block.

T_d delay time between rows (s).

P_d delay index, site specific.

Thus the block moves with an initial horizontal velocity V_h^d, V_r^c and vertical velocity V_z^c, without further acceleration, except due to gravity, until it lands on the muckpile or collides with other blocks in 'V' blast situations.

In-flight collisions. Although fragment collision occurs during all blasts, these can be neglected for blasts fired parallel to the free face. This statement is supported by analysis of high speed films of blasts. However, if 'V' initiation is used, in-flight collisions make an important contribution to the muckpile formation, because collisions reduce the energy, and slow the travel velocity of blocks. It is assumed that the most important collisions occur at the centre plane of a 'V' blast (defined in Fig. 1).

The model simulates effect of in-flight collisions using the Monte Carlo method. Whenever a fragment travels across the central plane of a 'V' blast (refer to Fig. 1), the total horizontal velocity of the block is multiplied by a random factor:

$$V_{hr}^{cls} = C_2 p V_{hr} \tag{19}$$

where p is randomly distributed between zero and 1.0 and C_2 is a constant, empirically taken to have a value of 0.9. V_{hr} is the total horizontal velocity before crossing the centre plane and V_{hr}^{cls}, the total horizontal velocity after crossing the centre plane.

Model input, output and capability

Model input

Input geometrical data includes hole collar coordinates and inclination, bench height, angle of free face, dip angle of floor and geometry of the boundary of the overburden. Explosive

input data includes the distribution of explosive in each hole (i.e. decking and stemming), initiation timing of each hole (used to define the detonating rows of holes). Grade region boundaries before the blast and the parameters P_e, P_r and P_d are also input.

Output of the model

A three dimensional muckpile shape and any cross section can be displayed. If information on grade regions before the blast is input to the model, a contour map of grade distribution within the muckpile can also be output.

Capability of the model

The model is capable of modelling most production blasts, which can be classified into the six types shown in Fig. 4:

Type 1: 'V' blast with side and front free face
Type 2: 'V' blast with only one front free face
Type 3: half 'V' blast with side and front free face
Type 4: half 'V' blast with only one front free face
Type 5: normal row by row blast with side and front free face
Type 6: normal row by row blast with only one front free face

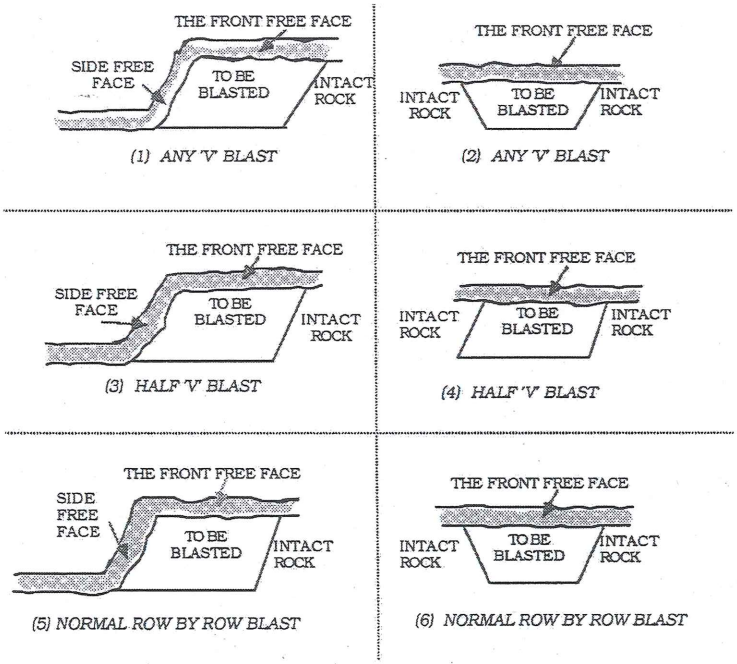


Fig. 4. Classification of production blasts that model can simulate

Various drilling patterns can also be modelled. Effects on the muckpile shape of any pre-blasted material adjacent to the overburden and other local geographical situations can also be simulated as obstacles to block displacement.

Model parameter calibration

Explosive field performance is determined not only by the strength properties of the explosive but also by the *in situ* rock mass condition. Effective coupling between the explosive and the rock also has a major influence on blasting results. The effects of explosive properties, rock mass conditions and interrow delays are represented in the model by control parameters, which are determined by calibration against field data. the criterion used to define 'best fit' is minimization of the function

$$F(P_e, P_r, P_d) = \frac{1}{N} \sum_{i=1}^{N} W_i (h^i_{obs} - h^i_{calc}(P_e, P_r, P_d))^2 \tag{20}$$

with respect to the parameters P_e, P_r and P_d. Here h^i_{obs} is the observed value of the muckpile profile at a point i, $h^i_{calc}(P_e, P_r, P_d)$ is the value calculated by the model for the given parameter values and W_i is a weight. A simple grid search method was used to find the minimum of the sum-of-squares function in Equation 20 (Yang and Kavetsky, 1989).

Case study results

At each field site, the model was first calibrated and then used in predictions and simulations to obtain result trends. Results were then compared over a range of designs. Knowledge about the sensitivity of changes to each of the blast design parameters reduced design errors and potentially increased the profits in the mining operation.

Simulation of different explosives used in the blast with the same hole diameter was achieved by making the model control parameter P_e relate to the explosive density (ρ), as follows:

$$\frac{P_e}{P_e(\text{for ANFO})} = \sqrt{\frac{\rho}{\rho_{ANFO}}} \tag{21}$$

This is based on the observation that in the model, P_e is proportional to the block velocity and the assumption that ρ is proportional to the energy per unit volume of an explosive. According to basic dimensional analysis, it is assumed that the velocity is proportional to the square-root of the energy. Therefore, P_e should be proportional to the square-root of ρ.

Coal overburden case study

The first case study was carried out based on data collected in May 1988 at a coal mine with a typical dragline/strip mining operation, located in central Queensland, Australia. At this site, overburden consists of sandstone with shale bands with a p-wave velocity between 1000 and 2000 m/s. The primary blasting objective is to achieve a loose, easily dug muckpile.

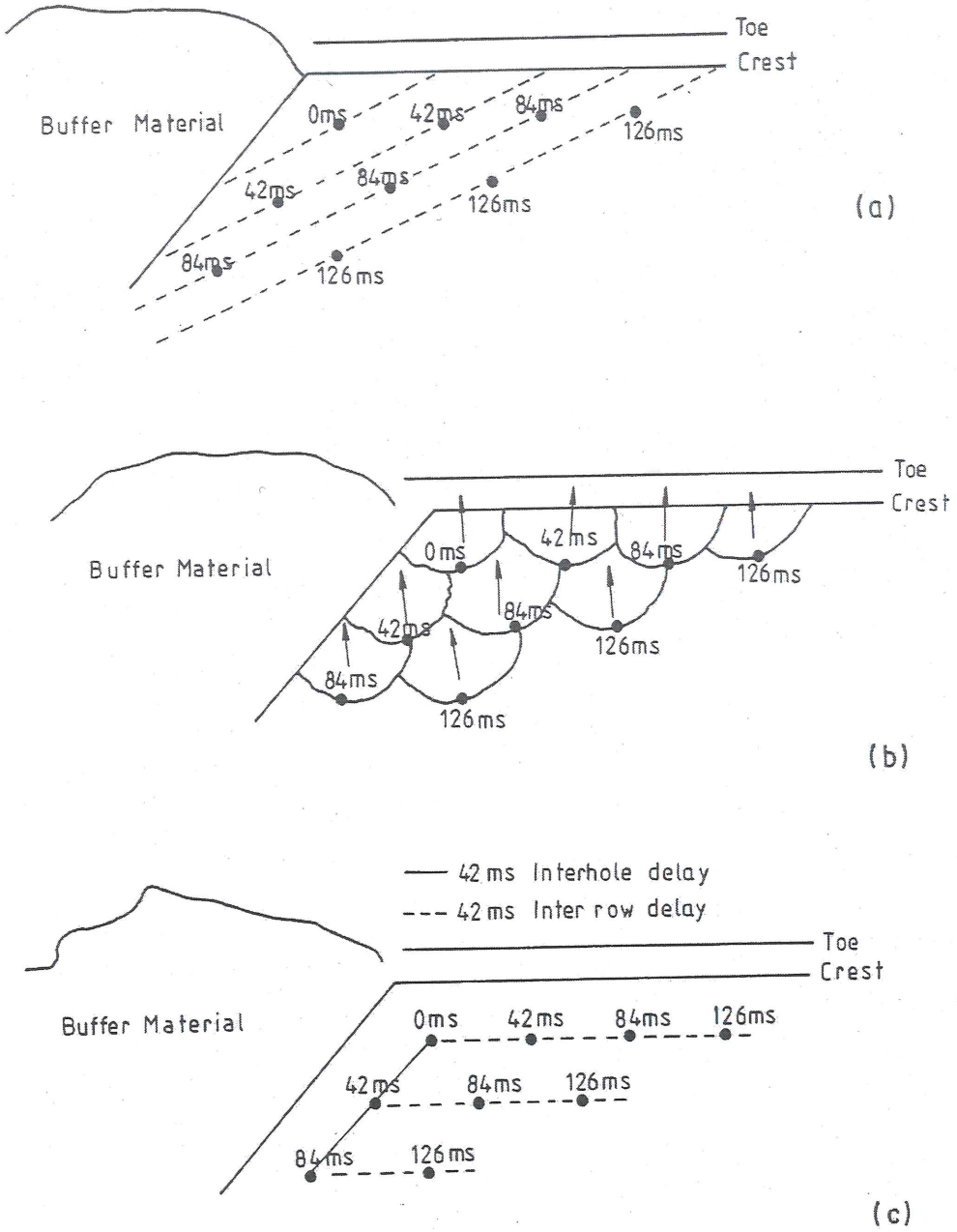


Fig. 8. Explanation of the burden movement process of Blast C, (a) It appears that the overburden would be divided in this way. This is not realistic because the holes with the same initiation time are located at a large separation distance. (b) Displacement process in practice, arrows showing the velocity direction of the material. (c) The blast can be modelled as a row by row parallel blast

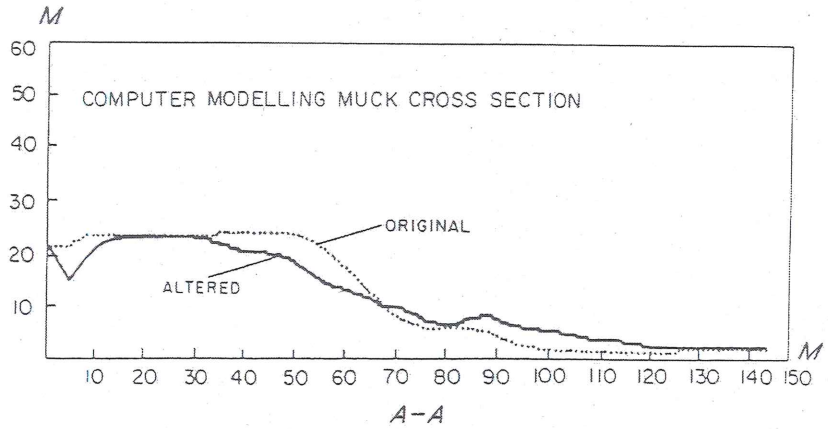


Fig. 9. Comparison between the predicted muckpile cross sections of Blast A; the original blast design: the initiation was 100 ms along the free face and 42 ms on the echelon and an altered blast design: the initiation was 42 ms along the free face and 100 ms on the echelon

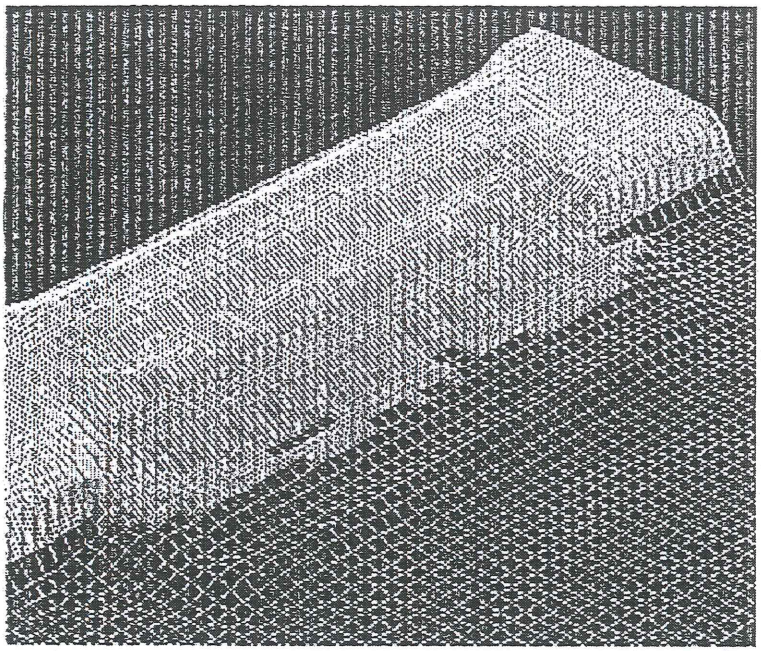


Fig. 10. A three dimensional display of the muckpile from the altered blast design for the overburden of Blast A

face, i.e. a Type 6 blast (refer to Fig. 4). A quite close data fit was obtained in the calibration. The calibrated model parameters derived were $P_e = 55$ for ANFO decks and $P_e = 69.6$ for HEF decks; $P_r = 1.35$ and $P_d = 0.01$.

Figure 11 illustrates the comparison between calculation of the grade (Al_2O_3) distribution (plan view) after Blast 1 and the grade distribution assayed before the blast. This demonstrates that most of the mixing occurred in the front part of the muckpile.

After the model parameters were calibrated, model was used to predict blast results. Blast 2 results were used to verify model predictions. The free face of the blast was irregular and was approximated in the simulation as a free face with toe and crest indicated in Fig. 12b. The blast was tied up in a 'V' pattern with 30 ms surface delays between the successive rows of the 'V' pattern. The centre region of the 'V' pattern was in the centre of the blast. However, a down hole delay of 200 ms between drilling rows parallel to the free face was also used. Therefore, the blast was basically still fired parallel to the free face. Consequently, the blast was still simulated as a normal row by row blast with only one front free face.The 200 ms was taken as an interrow delay and the inter-hole delay (30 ms) was neglected. Fig. 12a displays the predicted and surveyed muckpile cross sections. A quite good agreement is achieved. Figure 12b is the corresponding prediction of the grade (SiO_2) distribution (plan view) for Blast 2.

Blast 3 was initiated with a 'V' pattern containing interrow delays of 65 ms. Fig. 13a illustrates the surface tie-up of the holes. As indicated in Fig. 13a, it appeared that blast holes with the same initiation time would cooperate together as a detonation row to split the overburden on a plane passing through these holes and thus displace the rock in front the plane. However, the holes with the same initiation time were located a long distance apart. The overburden will be, more realistically, displaced in the way indicated in Fig. 13b. Therefore, Blast 3 was simulated as a Type 2 blast (refer to Fig. 4) with the initiation pattern in Fig. 13c, rather than the pattern as shown in Fig. 13a. Fig. 14 shows that the model prediction vs. the measured muckpile cross section provides quite an acceptable prediction.

In Fig. 15, a comparison is made between the surveyed muckpile A vs. the muckpile B as if the blast overburden and the charge condition of Blast 3 were initiated row by row parallel to the free face with a 65 ms inter-row delay, i.e. a Type 6 blast (refer to Fig. 4). It can be seen that increased throw would be achieved with this changed initiation for row by row detonation parallel to the free face compared to the original 'V' pattern.

In Fig. 15, simulated muckpiles (C and D) are from the blast designs with the same initiation sequence of row by row parallel to the free face (65 ms interrow delays), but with different decking designs. Muckpile C is predicted for the same blast overburden as that of Blast 3 with all charges being HEF. Muckpile D is the predicted muckpile when all charges in the blast are ANFO. From the figure, it is obvious that muckpile C is more displaced than muckpile B and that muckpile D is displaced less when compared with muckpile B.

Based on the above modelling, it was evident that blast initiation row by row parallel to the free face produced more throw than initiation with a 'V' pattern. This was because the dominant direction of the burden velocity was towards the free face in the parallel row initiation.

An important conclusion from the above simulations is that provided fragmentation is not affected, simplification of the current multi-deck explosive loading practice to a single column of ANFO would produce acceptable muckpile displacement.

Due to difficulties in collecting information on grade distribution after blasting, the predicted grade distribution has not yet been verified. If the prediction of grade distribution

Table 2. Blast Design Parameters of the Second Case Study

Design parameters	Blast 1	Blast 2	Blast 3
Free face angle	20°	20°	30°
Average bench height	18 m	11 m	18 m
Stemming	5.3 m	5.5 m	5.6 m
Dip angle of the bench floor	0°	0°	2°
Number of drilling rows	5	6	5
Staggered pattern B × S	9.6 × 9.4 m	9.4 × 9.6 m	9.6 × 9.4 m
Front row burden	3 m	3 m	5 m
Hole diameter	290 mm	290 mm	290 mm
Sub-drill	2–3 m	2–3 m	2–3 m

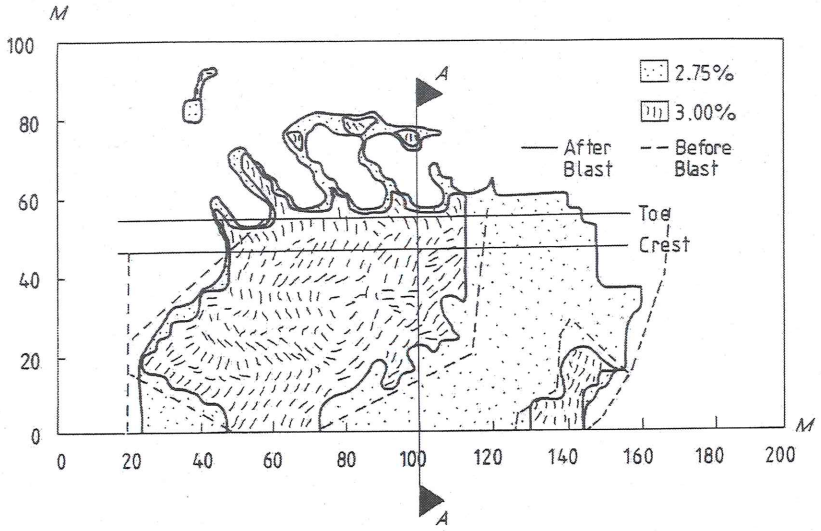


Fig. 11. Comparison between calculation of the grade (Al_2O_3) distribution (plan view) after Blast 1 and the grade distribution assayed before the blast

could be verified, such a predictive tool would enable blast design changes to be made to reduce the mixing of certain grade regions. At the very least, a good knowledge of post-blast region boundaries can assist in the layout of the grade boundaries and thus decrease grade discrepancies between the pit and the plant.

Discussion and conclusions

It was expected that the same interrow delay time at different field sites would have a different effect on the burden movement, because different rock masses would have a different response time to explosive loading. Therefore, the control parameter P_d was used to reflect

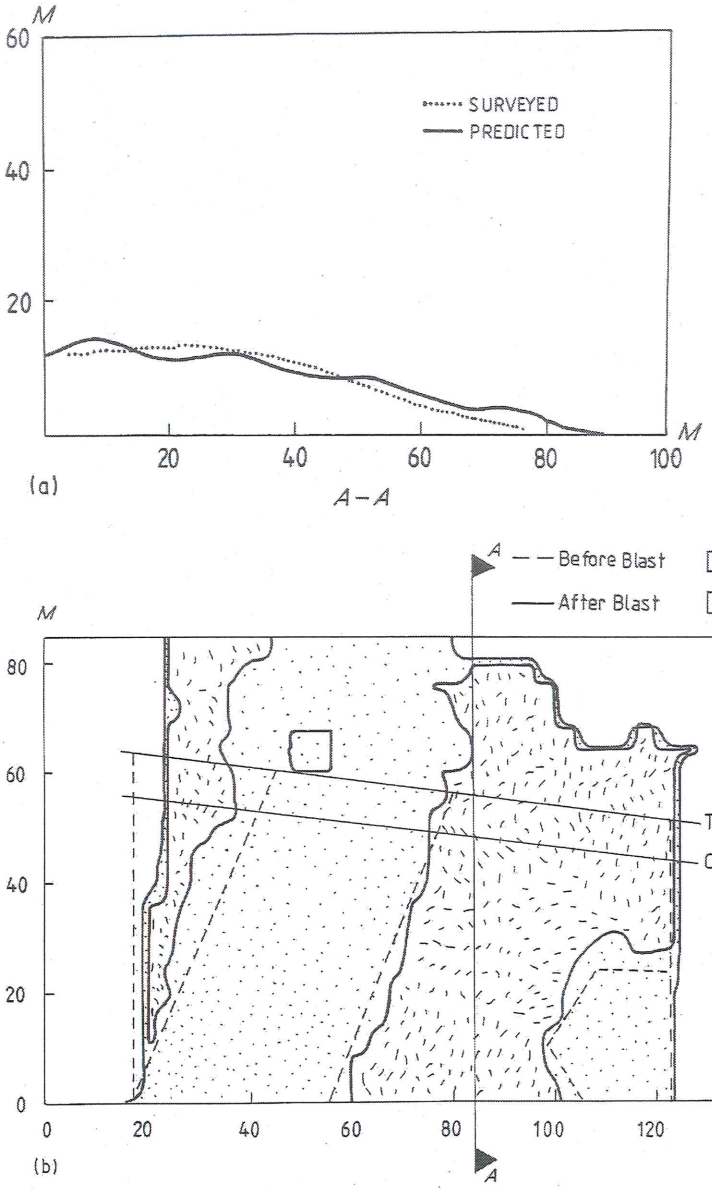


Fig. 12. Predictions for Blast 2, (a) Predicted muckpile vs. the surveyed for Blast 2; (b) Comparison between calculation of the grade (SiO_2) distribution (plan view) after Blast 2 and the grade distribution assayed before the blast

this site specific effect, as seen in Equation 18. However, from the above case studies, the calibrated value of P_d was constant for all of the above three sites. Further investigation is necessary to determine if P_d can be fixed as a constant so that the model will have only two controlling parameters (P_e and P_r), which need to be calibrated at any new site.

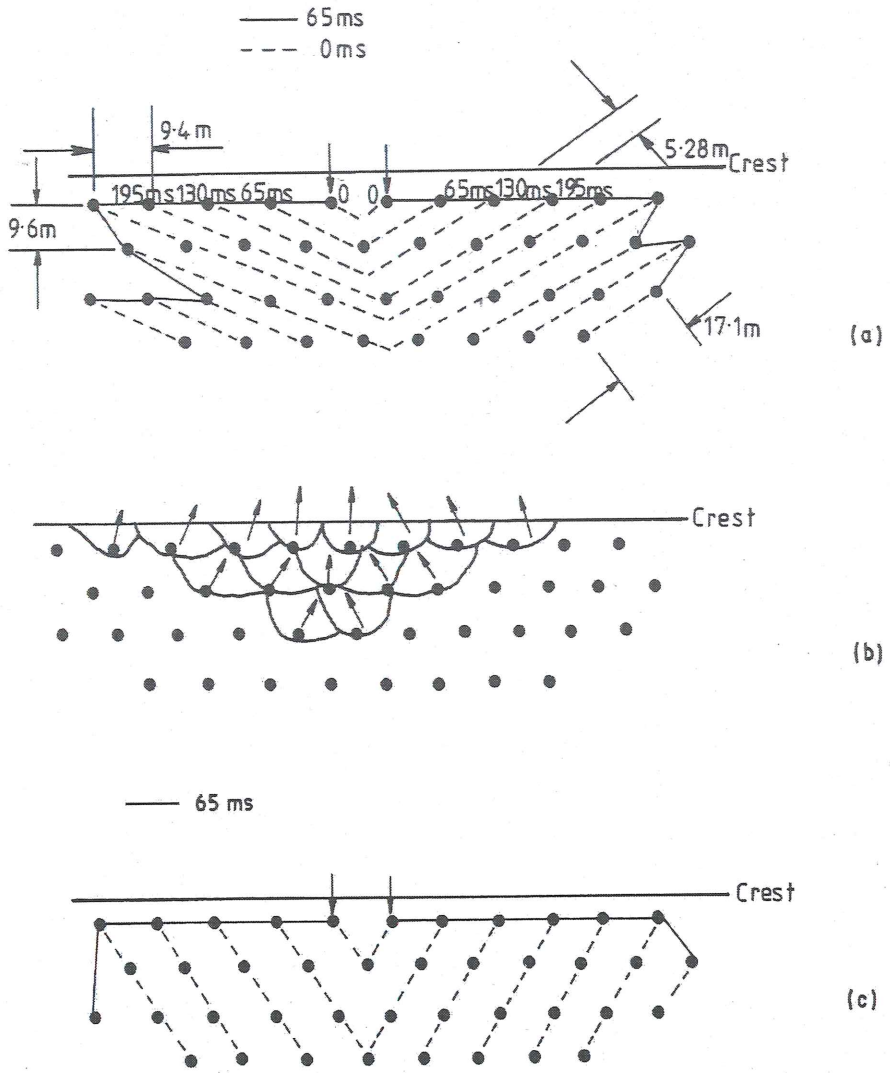


Fig. 13. Explanation of the burden movement process of Blast 3. (a) It appears that the overburden would be divided in this way. This is not realistic because the holes with the same initiation time are located at a large separation distance (b) Displacement process in practice, arrows showing the velocity direction of the material. (c) The blast can be modelled this 'V' pattern blast

Due to difficulties in collecting data covering all the different types of production blasts that the model can simulate, the case studies above have not yet fully demonstrated the capacity of the model to simulate different blast patterns. Future work will be required to apply the model to many different sites and as many blasts as possible at the same site. In this way, the model can be tested further and also developed further.

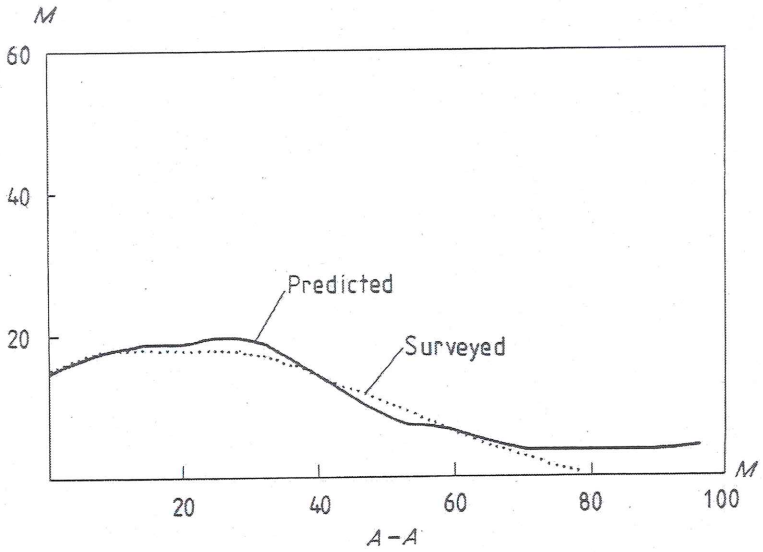


Fig. 14. Predicted muckpile vs. the surveyed for Blast 3

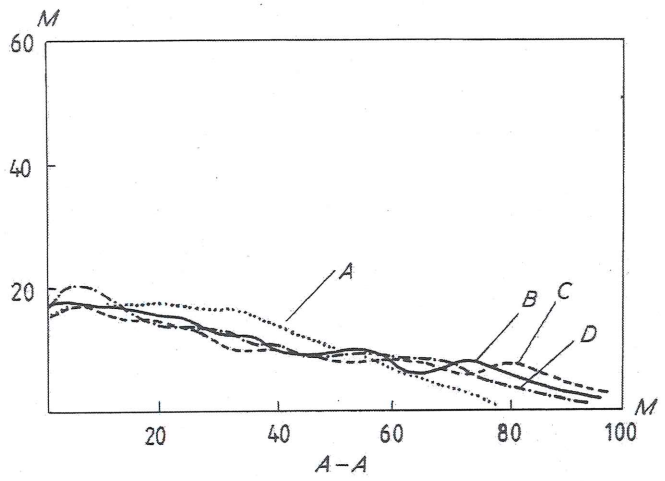


Fig. 15. Comparison between the surveyed and predicted muckpiles. A—the surveyed after Blast 3. (B, C and D)—predicted with different explosive charge conditions if the overburden of Blast 3 is initiated row by row parallel to the free face with 65 ms inter-row delays: B—with the same charge condition as that of Blast 3; C—with all charges being HEF; D—when all charges in the blast are ANFO.

Some limitations of the model exist at this stage. Before the simulation of a production blast, the effective firing rows of the blast have to be defined. The interrow delay time is simulated explicitly. As is seen in the above case studies, the model can not take the inter-hole delay directly into account. After definition of the detonation rows, inter-hole delay is

neglected. In future developments of the model, an inter-hole delay should be considered. This may improve predictions for 'V' pattern blasts.

The 3D-kinematic model presented in this paper is capable of predicting muckpile shape and grade boundary movement. These factors are of great importance to mining efficiency in open pit mines.

The model is based on a simple 'kinematic' approach to model muckpile formation, which includes both fundamental and empirical aspects. Because the explosive properties and rock mass condition are very difficult or impossible to measure in terms of blast performance these are simply represented by control parameters which are calibrated in the field.

Because of its relative simplicity, the model is simple to calibrate at any site and easy to implement in simulations. The geometery of the blast design has a very important effect on blasting performance. The model presented in the paper is capable of modelling most of the productions blasts with accurate geometry calculations.

Case studies of model applications have demonstrated that the model can predict reasonable trends, results and comparisons for a range of designs. Consequently, optimizing blast designs can be significantly enhanced by utilizing the full capabilities of the model.

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